exists. Notice that 2^n has first digit k for some $k \in \{1, 2, \dots, 9\}$ if and only if

$$\log_{10} k \leqslant \{n \log_{10} 2\} < \log_{10}(k+1),$$

where we write $\{t\}$ for the fractional part of the real number t.

Since $\alpha = \log_{10} 2$ is irrational, we may apply (1.2) to deduce that

$$\begin{aligned}\frac{\left|\{n \,|\, 0 \leqslant n \leqslant N-1, \, 1^{\text{st}} \text{ digit of } 2^n \text{ is } k\}\right|}{N} &= \frac{1}{N}\sum_{n=0}^{N-1} \chi_{[\log_{10} k, \log_{10}(k+1))}(R_\alpha^n(0)) \\ &\to \log_{10}\left(\frac{k+1}{k}\right)\end{aligned}$$

as $N \to \infty$.

Thus the first digit $k \in \{1, \dots, 9\}$ appears with density $\log_{10}\left(\frac{k+1}{k}\right)$, and it follows in particular that the digit 1 is the most common leading digit in the sequence of powers of 2.

Exercises for Sect. 1.1

Exercise 1.1.1. A point $x \in X$ is said to be *periodic* for the map $T : X \to X$ if there is some $k \geqslant 1$ with $T^k(x) = x$, and *pre-periodic* if the orbit of x under T is finite. Describe the periodic points and the pre-periodic points for the map $x \mapsto 10x \pmod 1$ from Example 1.1.

Exercise 1.1.2. Prove that the orbit of any point $x \in \mathbb{T}$ under the map R_α on $\mathbb{T}$ for α irrational is dense (that is, for any $\varepsilon > 0$ and $t \in \mathbb{T}$ there is some $k \in \mathbb{N}$ for which $T^k x$ lies within ε of t). Deduce that for any finite block of decimal digits, there is some power of 2 that begins with that block of digits.

1.2 Equidistribution for Polynomials

A sequence $(a_n)_{n\in\mathbb{N}}$ of numbers in $[0,1)$ is said to be equidistributed if

$$\mathbf{d}(\{n \in \mathbb{N} \mid a \leqslant a_n < b\}) = b - a$$

for all a, b with $0 \leqslant a < b \leqslant 1$. A classical result of Weyl [381] extends the equidistribution of the numbers $(n\alpha)_{n\in\mathbb{N}}$ modulo 1 for irrational α to the values of any polynomial with an irrational coefficient*.

* Numbered theorems like Theorem 1.4 in the main text are proved in this volume, but not necessarily in the chapter in which they first appear.